

Multivariate regression & model selection

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From uni- to multi-variate

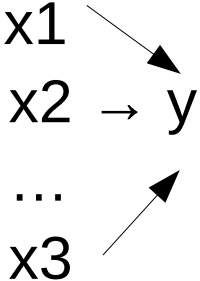
- Up to now:

$$x \rightarrow y$$

$$\hat{y} = b + a * x$$

or e.g.

$$\hat{y} = \exp(b + a * x)$$

- New: 

A diagram illustrating a multi-variate model. On the left, the variables x_1 , x_2 , an ellipsis ($\dots$), and x_3 are listed vertically. On the right is the variable y . Arrows point from x_1 and x_3 towards y , while a direct arrow points from x_2 to y .

From uni- to multi-variate

$$\hat{y} = b + a_1 * x_1 + a_2 * x_2 + \dots + a_n * x_n$$

-
- How large “n” is depends on the number of variables available to you... and on your decision what variables to use

From uni- to multi-variate

$$\hat{y} = b + a_1 * x_1 + a_2 * x_2$$

- vs

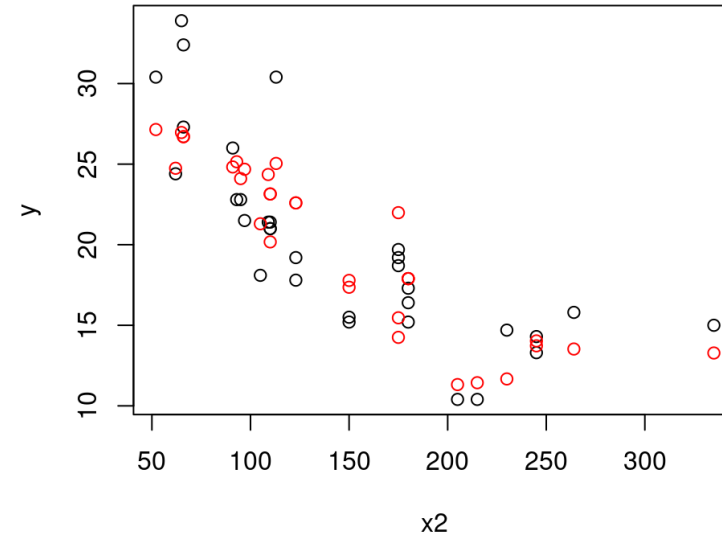
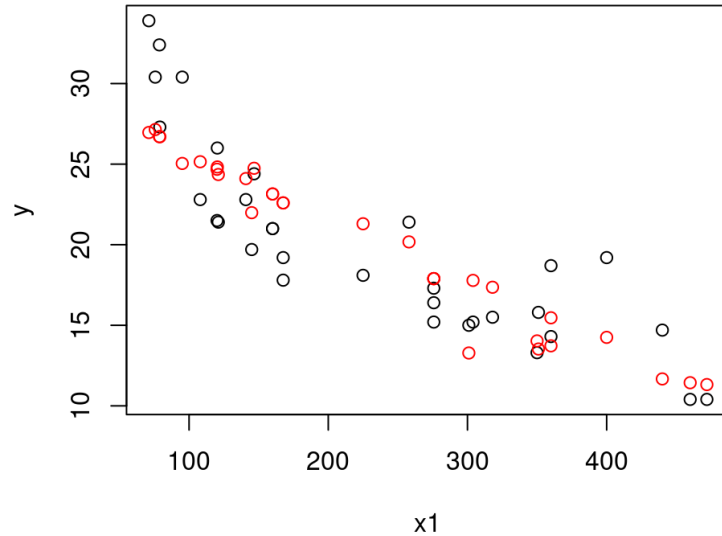
$$\hat{y} = b + a_1 * x_1 + a_2 * x_2 + a_3 * x_1 * x_2$$

- This means that the effect of x_1 on y changes as x_2 increases / decreases, and vice versa
- = “interaction term”

GLM function formulation

- `glm(y ~ x1 + x2)`
-
- `glm(y ~ x1 + x2 + x1:x2)`
- alternatively: `glm(y ~ x1 * x2)`

Partial effects

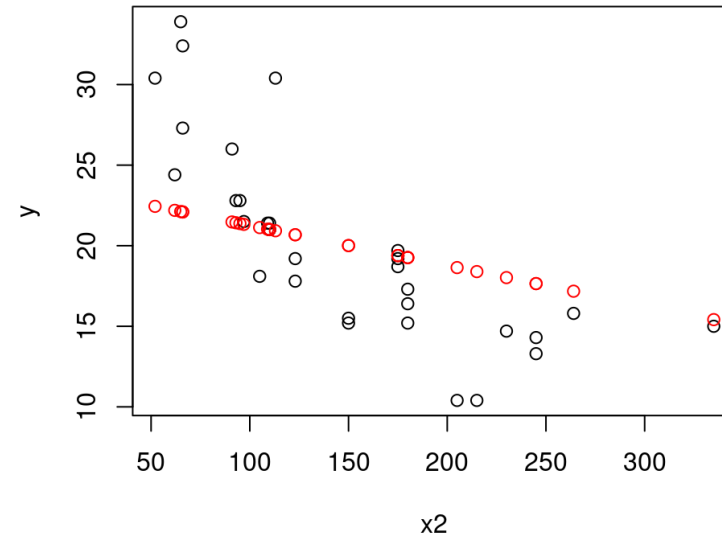
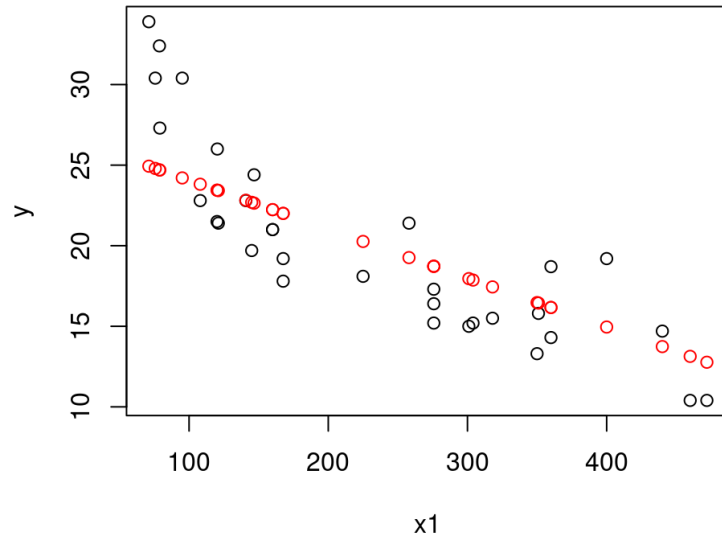


- these are not partial effects
- these are predictions for the “ x_1 - x_2 pairs” of fitting data, plotted separately against x_1 and x_2

Partial effects

- We want to know the isolated effects of x_1 and x_2 on y
- Means, for partial effect of x_1 , we integrate over x_2
- For a GLM, we can simply make a prediction with x_2 set to the average of x_2 in the fitting data
- `predict(model, newdata = list('x1' =
x1, 'x2' = rep(mean(x2), length(x2))))`

Partial effects



- x_1 appears to have a stronger partial effect on y than x_2
- both partial effects are negative
- in general it has greater relevance in more complex models (GAMs, regression trees...)

How to decide on variables and interactions

- Prior information / choice / task
- → what variables make sense to consider (from a theoretical POV / through expert knowledge or consulting literature)?

How to decide on variables and interactions

- → what variables are actually available (= what was measured, what can you obtain from literature / online resources)? (Can you actually address your research question with the data available?)
- → what variables are (likely) correlated with one another? (e.g. salinity and conductivity)

How to decide on variables and interactions

- While programming / actively modeling:
- → what combination of variables and / or interactions result in the best model quality

Assessing model quality - criteria

- Not: the p-values
-
- Variance explained: R^2 (the higher, the better)
- Akaike Information Criterion (AIC): basically speaking, weighs R^2 against the number of parameters (the lower, the better)
- Diagnostics plots (mainly spread of residuals)

Why p-values are the wrong way to judge your model

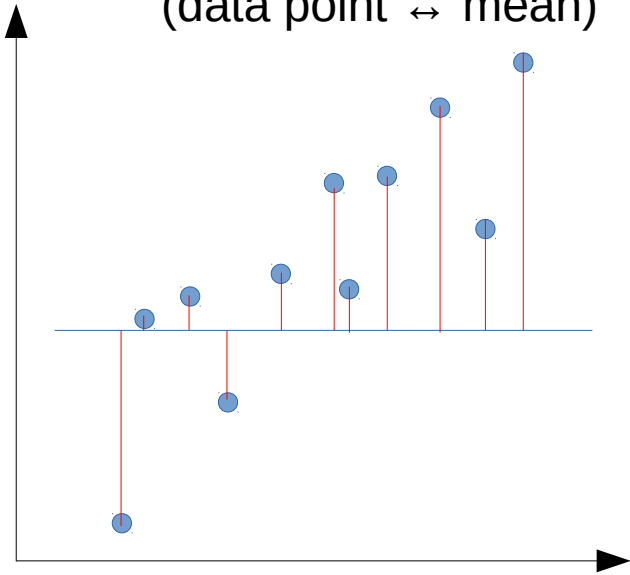
- Attaining significance (a p-value ≤ 0.05) is a scientist's career aim (especially if you work in empirical science, e.g. field work or clinical science)
- You can get a low p-value with a poorly-configured model, but then it is practically worthless

Why p-values are the wrong way to judge your model

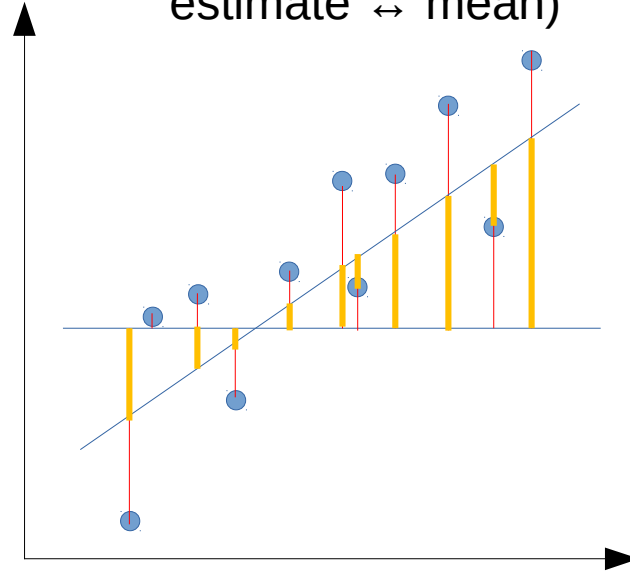
- A good model should be a good representation of “reality” (or of the parts of reality that we perceive through measurements) first and foremost!
- A good model is not one that explains the variation in the y-variable very well... or one that delivers exciting results
- (a good model *may* do these things, but it may just as well not)

R^2

“empirical variance”
(data point $\leftrightarrow$ mean)



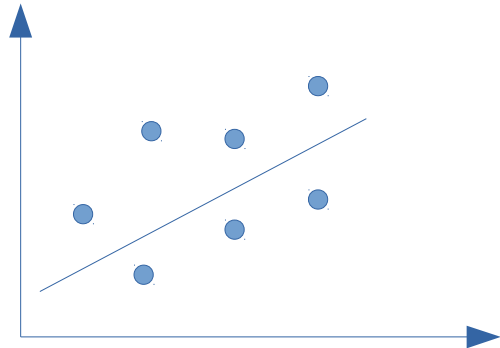
“model variance” (regression
estimate $\leftrightarrow$ mean)



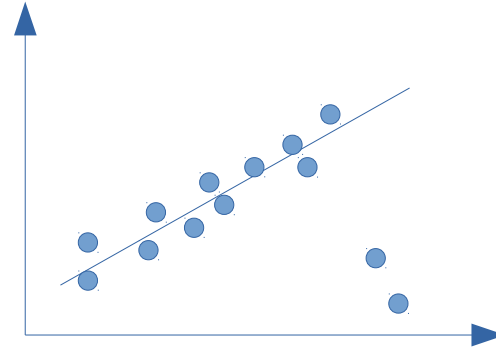
$$R^2 = \text{summed "model variance"} / \text{summed "empirical variance"}$$

Model diagnostics

- Explained variance can't stand alone → always back it up with additional quality checks
- Why? Explained variance is an aggregate measure. Imagine...



- rather low explained variance
- (probably no significant effect)
- good model!

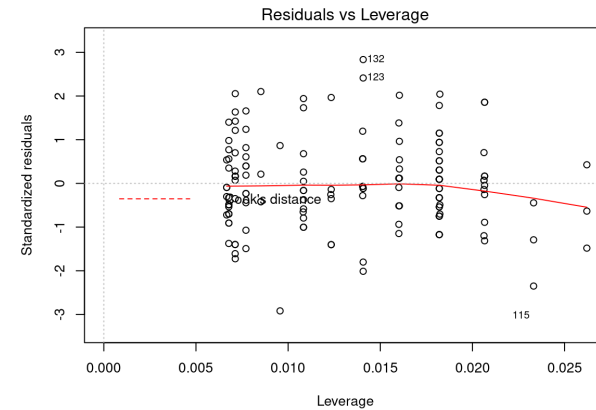
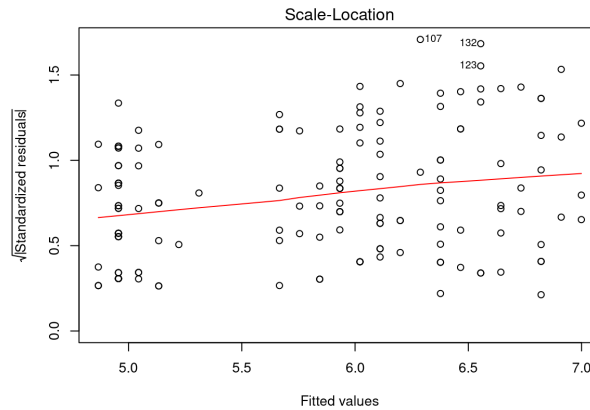
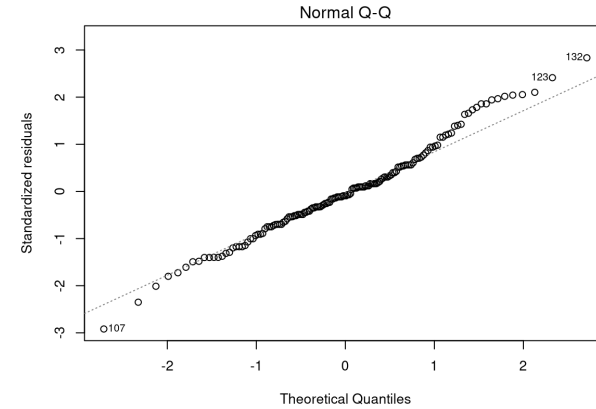
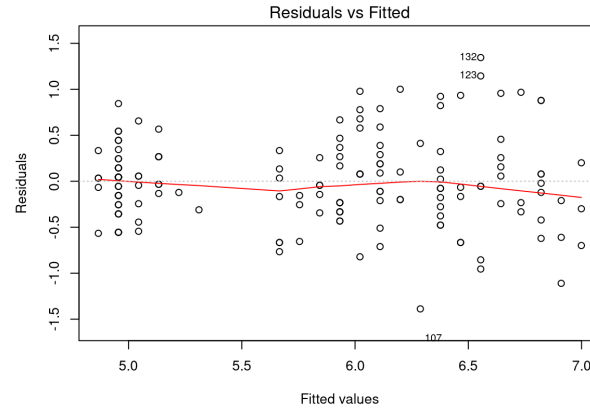


- rather high explained variance
- poor model!

Model diagnostics

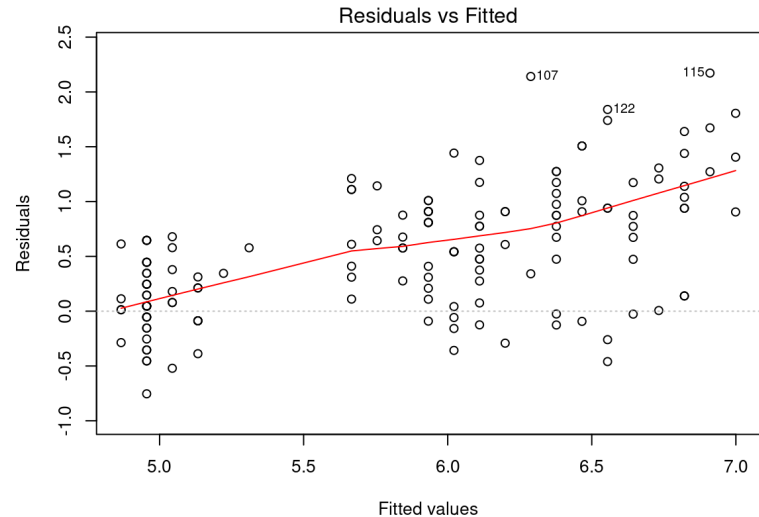
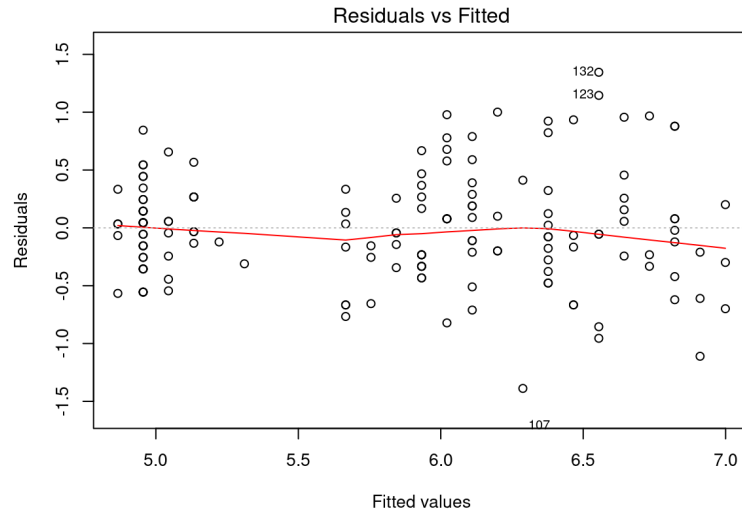
- The crucial part is having the same degree of explained variance over the full range of measured / observed y
- Both in terms of residuals directions (should be heterogeneous) and magnitudes (should ideally not change over the x -axis, but a bit less important than not observing patterns in the directions of the residuals)
- Assessing diagnostics plots requires some experience, especially when performance is similar between two models

Diagnostics plots



Diagnostics plots – upper left

- Shows the residuals ($y_{\text{predicted}}$ minus y_{observed}) against $y_{\text{predicted}}$
- Ideally **no trend** over the x-axis
- Ideally the residual spread should not change over the x-axis

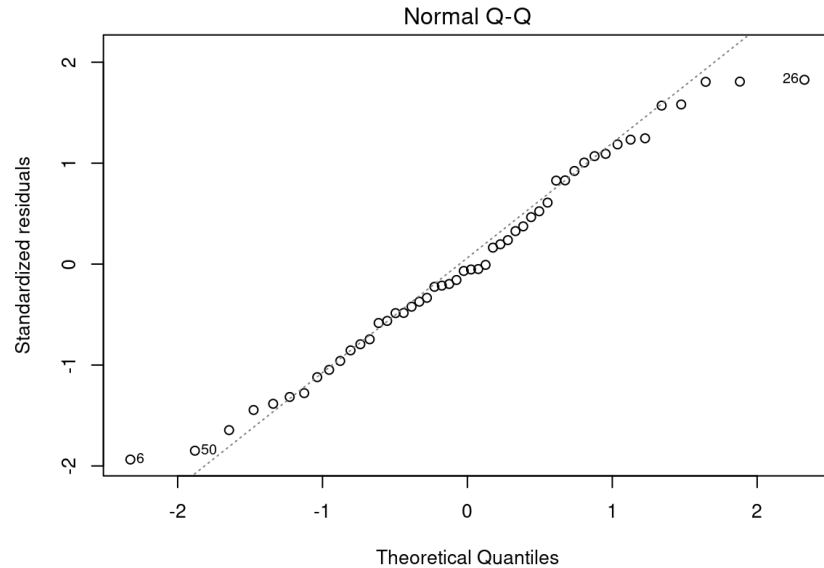


Diagnostics plots – upper right

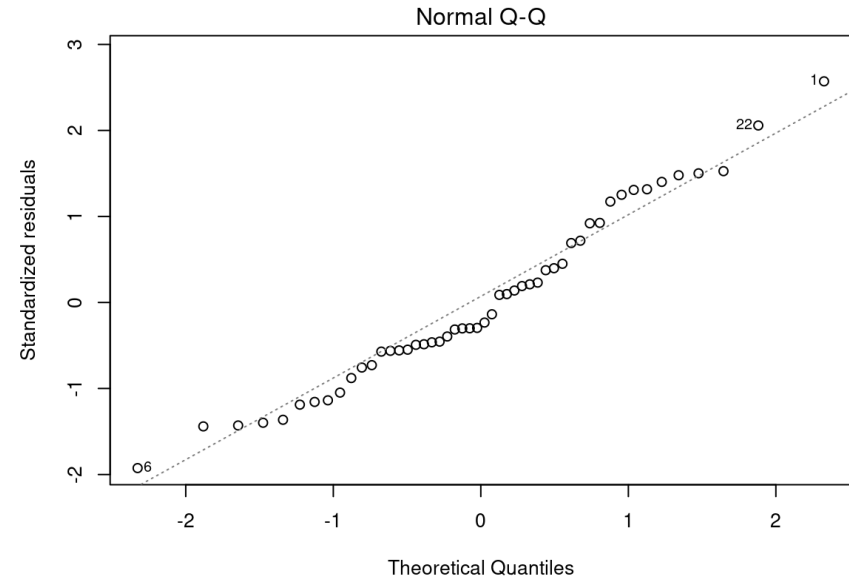
- “Q-Q” plot or “quantile-quantile” plot
- → actual (standardized) residuals plotted against the quantiles that you would expect if the spread of the residual “data points” over the residual value range was perfectly uniform (?)
- Ideally all data points should lie on the “equality line” (slope = 1)

Diagnostics plots – upper right

better



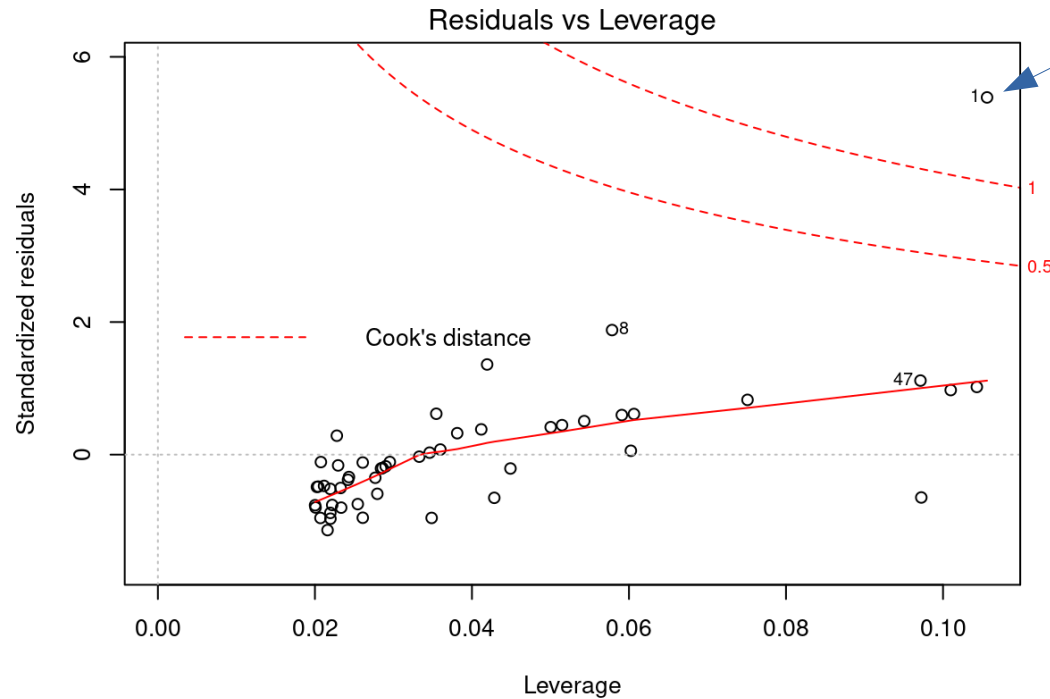
not so good



Diagnostics plots – lower right

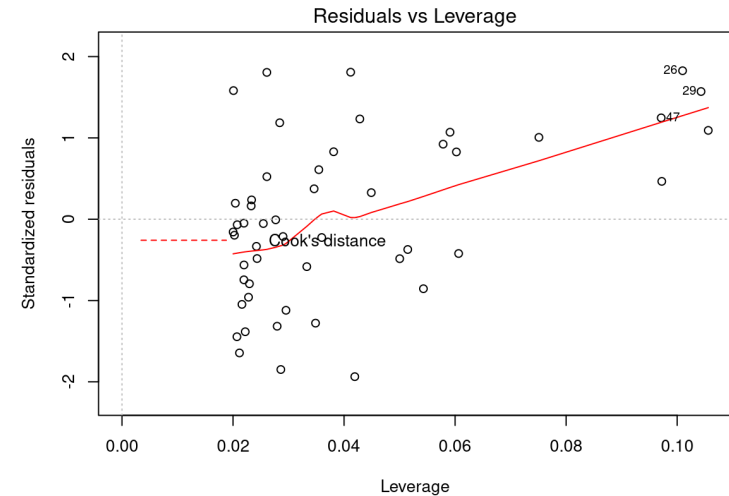
- Gives you an idea of potential outliers
- Basically tells you about the impact of single data points on the overall model fit
- Normally, you wouldn't like to have data points that have a particularly large impact compared to the rest
- If you forget to log-transform log-normally-distributed data, this will be a typical case

Diagnostics plots – lower right



Avoid having data points “beyond” these dashed lines

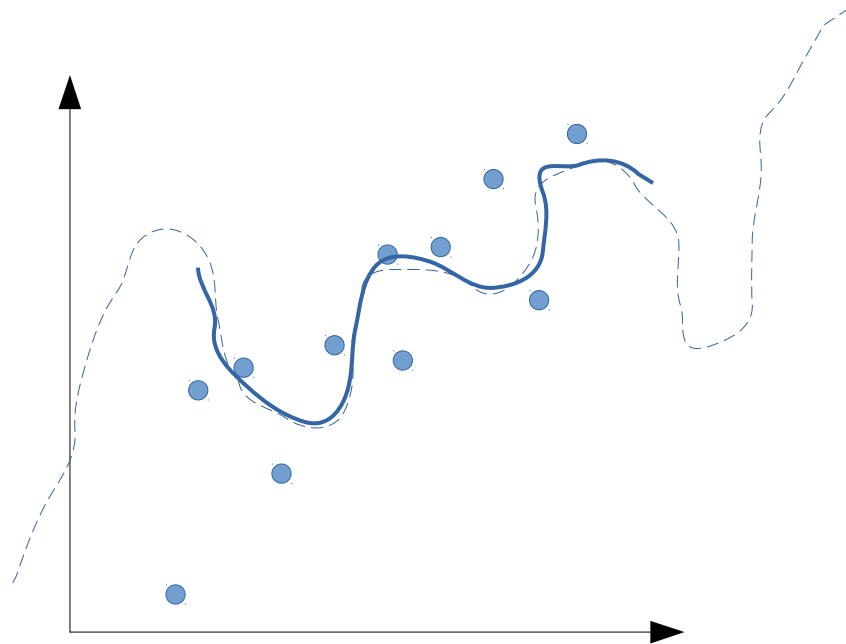
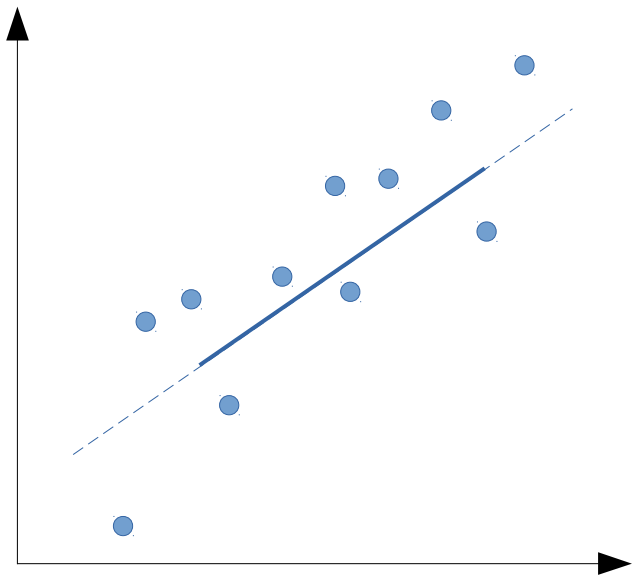
This is better:



Akaike Information Criterion (AIC)

- “Balanced” metric accounting for R^2 and the number of model parameters
- A high R^2 is often no good if the number of parameters is (very) high → models tend to overfit
- Principle of “Occam’s razor”: the simplest model is often the best
- → i.e., the model with lower AIC is often the better one
- When noise is rather high, this principle might however select the simpler model for the wrong reason
→ extrapolation is then very dangerous

Akaike Information Criterion (AIC)



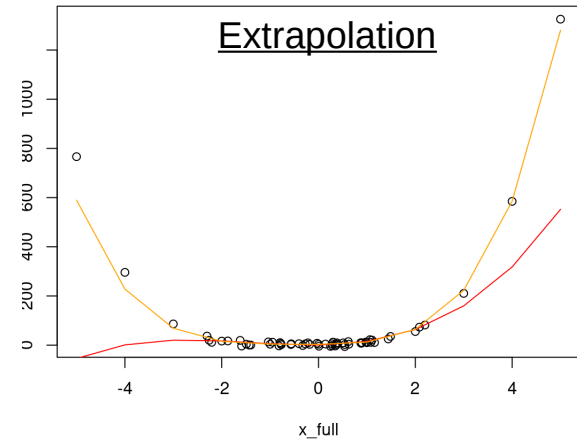
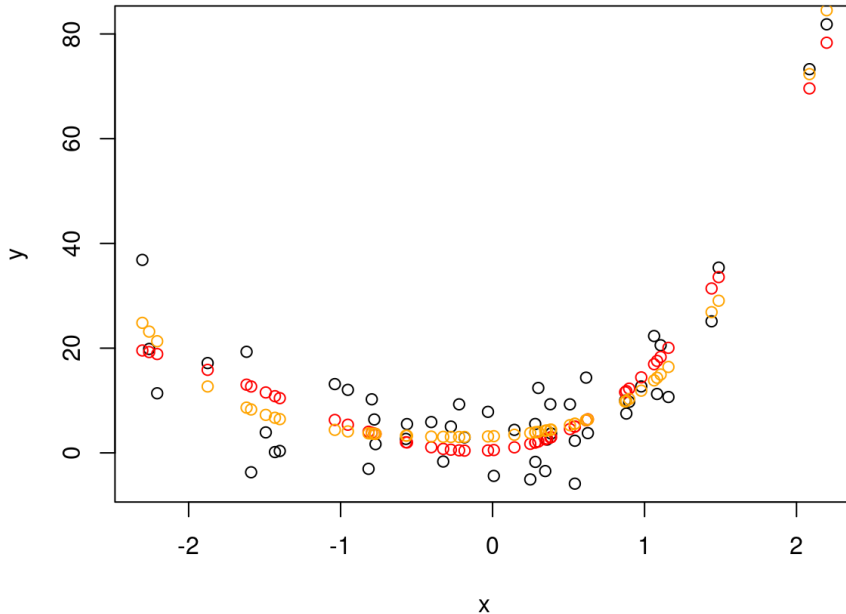
Akaike Information Criterion (AIC)

y-values are fourth-order polynomial of x plus noise drawn from normal distribution with SD = 8

Red dots: fit of a third-order polynomial
Orange dots: fit of a fourth-order polynomial
(= the correct but more complex model)

AIC for red dots: 339

AIC for orange dots: 331



Akaike Information Criterion (AIC)

- If you have two candidate models, the preferred one (in terms of AIC) is the one with lower AIC
- Rule: a “meaningful” difference is a difference of ≥ 2 (e.g. 300 vs 302)
- If you want to compare AICs, the y-values need to be equal! Don’t compare models fitted on different subsets of data, and don’t compare transformed and non-transformed y-values!

Ockham's razor

- Generally, when in doubt, go for the simpler model
- The AIC also promotes this, advocating for a model with fewer parameters
- But it also gives you a guideline if difference between AICs is < 2
- Choosing simpler model generally means making fewer assumptions

How to decide on variables and interactions

- Backwards selection: you start with the most complex model: All variables* and all conceivable interactions
- You drop the most complex interaction, and compare the R^2 , the AIC and the diagnostics plots to the prior model
- One by one, you drop interactions, and then single variables, while monitoring the quality-control metrics

How to decide on variables and interactions

- You stop when any of the quality-control metrics worsens again (i.e. the model becomes too simple at that point)
- If multiple configurations with the same degree of complexity are possible (e.g. 3 two-variable interactions), you test all three of them at the same step

How to decide on variables and interactions

- You don't drop a single variable before having dropped an interaction containing that variable
- (i.e., don't set up a model like this:

lm(y ~ x1 + x3 + x1:x2 + x2:x3 + x1:x3))

- Also, in case of more complex interactions (> 2 predictor variables): don't drop a simpler interaction if that interaction is included in your more complex interaction

How to decide on variables and interactions

- Backward selection is quite tedious
- Maybe it could be automatized with AI agents? But no 100 % certainty...
- Also it becomes basically undoable when you have a large number of variables... the number of interactions increases non-linearly with the number of variables

How to decide on variables and interactions

- Alternative: forward selection
- → start with a single variable and add others, and interactions, step-by-step
-
- Often, it is useful to make use of *prior* knowledge for selecting variables and interactions (= “what makes sense?”, “what is already reported in the literature?”)

Demonstration model selection

Exercise model selection

- 1) you get a multivariate data set. Select the correct model via backward selection (each on their own). Plot the partial effects of the predictor variables on a model of your choice (final selection or not)
- 2) in groups (of three) create your own data set ($x_1, \dots, x_n, y$, plus random noise). Then swap among groups. Then identify the correct model via backward selection

Exercise model selection

- We don't address data transformation today
- (except if you include it in your custom-created datasets ;-))